NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-22885/96B(EC)
27 September 1991

SUPERSEDING
MIL-S-22885/96A(EC)
14 September 1982

Ⓑ PERFORMANCE SPECIFICATION SHEET

SWITCH, PUSHBUTTON, ILLUMINATED, 4-LAMP, RFI SHIELDED

This specification sheet is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-22885.

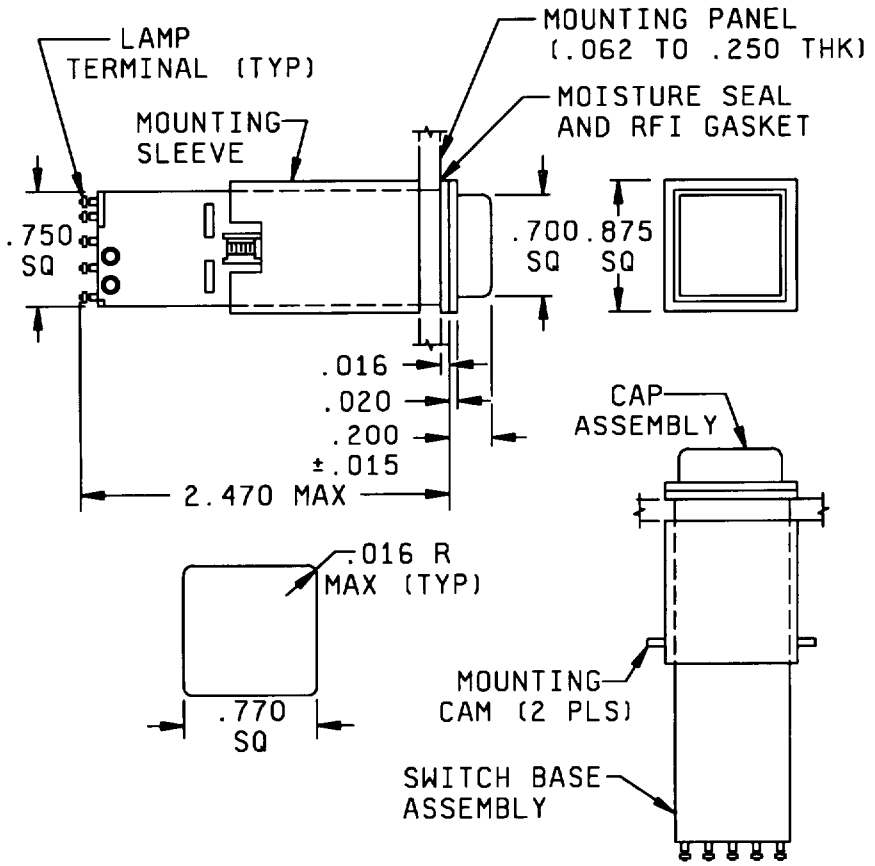
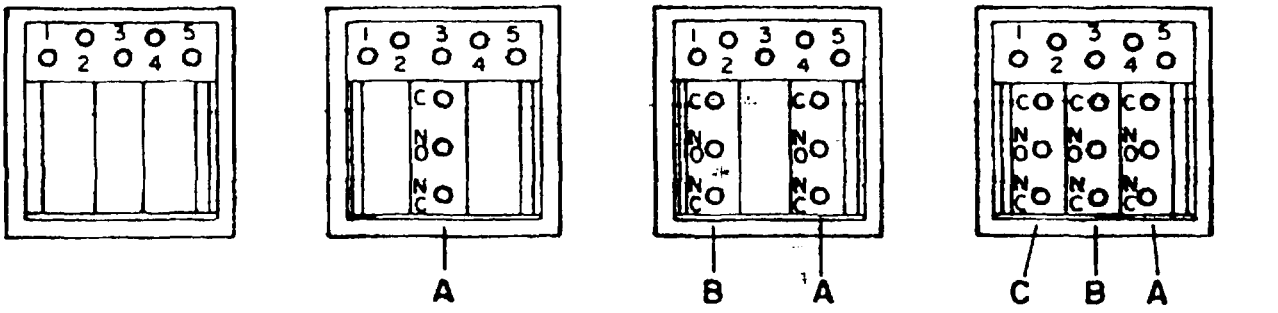


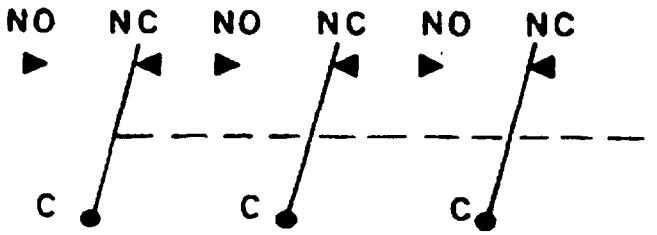
FIGURE 1. Design and construction.

Ⓑ denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:32Z
Check the source to verify that this is the current version before use.



TERMINAL IDENTIFICATION



SWITCH CIRCUITRY
3PDT SHOWN

Inches	mm
.015	0.38
.016	0.41
.020	0.51
.062	1.57
.200	5.08
.250	6.35
.700	17.78
.750	19.05
.770	19.56
.875	22.22
2.470	62.74

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (25 mm)
4. Basic switch shall be MS24547-4 or MS24547-5 (MIL-S-8805/4).
5. Lens holder can slide off when cap is removed from base. Each cap assembly shall accommodate a film legend of .645 x .645 ± 010 x 004 ± 001 (16.38 x 16.38 ± 25 x 10 ±.03 mm) not supplied (available legend area .800 sq).
6. Cap assemblies shall be held captive to base assemblies by retaining element to prevent accidental interchange
7. Lamp terminals shall be permanently identified as shown. Switch terminals shall be identified as "C", "NC" and "NO"

FIGURE 1. Design and construction - Continued.

Source: https://assist.dla.mil - Downloaded: 2026-10-19 14:32Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Dimensions and configurations See figure 1

Complete switch shall consist of

Base assembly Includes basic switches (MS24547-4 or -5), mounting sleeve and moisture /RFI gasket.

Cap assembly Includes necessary color filters, diffusers and front lens plate (.600 square lens engraving area) Accepts 4 T-1-3/4 midget flange base incandescent lamps, not included, order separately Accommodates a film legend of .645 x .645 ±.010 x .004 ± 001, not supplied. Includes RFI screen

(B) Enclosure design 2 (dripproof) panel seal only. Cap assembly and lamp circuit not sealed.

Temperature characteristic: 1 (-55°C to +85°C)

Vibration grade 2 (10-500 Hz)

(B) Dielectric withstanding voltage: Dielectric withstanding voltage on lamp circuit not applicable after seal test.

Operation force: 40 ounces maximum

Cap assembly extraction force: 4 ±1 pounds

Lamps: Accommodates 4 T-1-3/4 midget flange base lamps. Not included, order separately. Select lamps of 75 watt or less power rating

(B) Display type. W, translucent white, which when illuminated shows the required color

Material: Base assembly and mounting sleeve. Stainless steel.

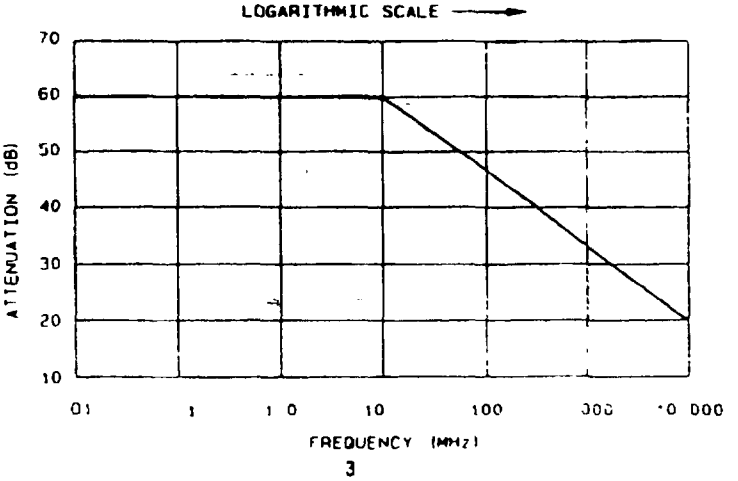
Electrical ratings: See tables I and II

Luminance and color See table III

Weight: .12 pound maximum

(B) EMI/RFI shielding. Cap-lampholder assembly to have screen and contact for RFI shielding. Switches shall be tested to determine shielding effectiveness in accordance with the shielding efficiency test of MIL-S-22885 The test shall consist of electric field and plane wave measurements performed at the following frequencies Attenuation shall not be less than specified below

Electric field	Minimum attenuation	Plane wave	Minimum attenuation
15 kHz	60 dB	100 MHz	47 dB
150 kHz	60 dB	400 MHz	39 dB
500 kHz	60 dB	1 GHz	33 dB
1 MHz	60 dB	2.5 GHz	28 dB
10 MHz	60 dB	10 GHz	20 dB



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:32Z
Check the source to verify that this is the current version before use.

TABLE I Electrical ratings (silver contacts)

Load	Sea level		50,000 feet
	28 V dc (amperes)	115 V ac, 60 Hz (amperes)	20 V dc (amperes)
Resistive	7	7	4
Inductive	4	7	2 5
Lamp	2.5	2	2 5

TABLE II. Electrical ratings (gold contacts)

Load	Sea Level 28 V dc (amperes)	50,000 feet 28 V dc (amperes)
Resistive	1	1
Inductive	.5	.5

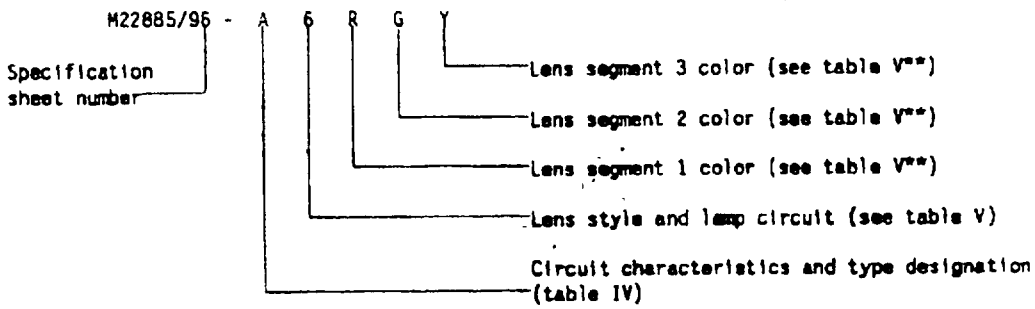
TABLE III Luminance and color.

Color	Chromaticity limits @ 2,360 Kelvin		Luminance 1/ (footlamberts) minimum average
	X	Y	
Green (G)	.260 .325 .160 .225	570 SL 2/ .680 SL 2/	40
Red (R)	695 703 655 660	285 SL 2/ 325 SL 2/	60
White (W)	400 460 460 400	460 460 .380 380	40
Yellow (Y)	562 570 596 605	.415 SL 2/ .382 SL 2/	150
Blue (B)	150 220 150 220	420 420 .320 320	20

1/When illuminated by two MS25237-327 lamps of 0 34 ± 02 mean spherical candle power or by four lamps of 15 ± 02 mean spherical candle power with a maximum power output of 2 5 watts
2/SL - Spectrum locus (where intersected by other coordinate pair)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:32Z
Check the source to verify that this is the current version before use.

Part or identifying number (PIN). PINs shall be assigned as illustrated below:



** Leave blank when not applicable

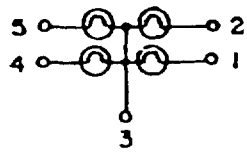
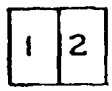
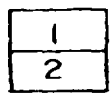
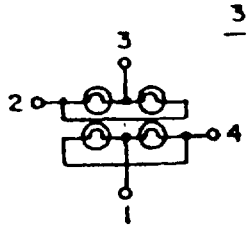
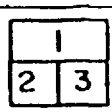
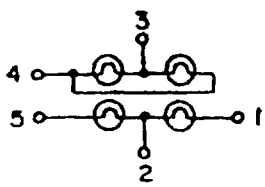
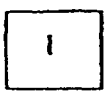
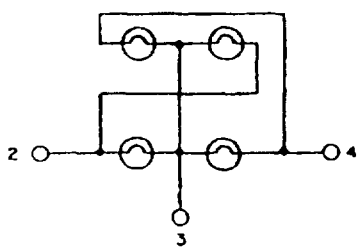
TABLE IV Circuit characteristics and type designation

Code 1/	Contact material	Circuit characteristics	Switch action
U	Silver	1 PDT	Momentary
A	Silver	2 PDT	Momentary
V	Silver	3 PDT	Momentary
W	Gold	1 PDT	Momentary
C	Gold	2 PDT	Momentary
Y	Gold	3 PDT	Momentary
Z	Silver	1 PDT	Alternate
J	Silver	2 PDT	Alternate
1	Silver	3 PDT	Alternate
2	Gold	1 PDT	Alternate
L	Gold	2 PDT	Alternate
3	Gold	3 PDT	Alternate
4	Silver	1 PDT	Alternate holddown
5	Silver	2 PDT	Alternate holddown
6	Silver	3 PDT	Alternate holddown
7	Gold	1 PDT	Alternate holddown
8	Gold	2 PDT	Alternate holddown
9	Gold	3 PDT	Alternate holddown
S	None	None	Indicator
X 2/	None	None	---

1/Code letters B, D, E, F, G, H, K, M, N, P, Q, R, and T for 4PDT and plug-in module type switches have been canceled
2/Use code X to specify cap assembly portion of switch only

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:32Z
Check the source to verify that this is the current version before use.

TABLE V. Lens style and lamp circuit

CODE 1/	LENS STYLE 2/	LAMP CIRCUIT
1		
2		
3	None - switch body only	
4		3/ 
5	None - switch body only	
6		
7	None - switch body only	
8		
9	None - switch body only	

1/ Previously assigned code letters A, B, and C have been superseded by 1, 4, and 6, respectively. Code letter D has been superseded by 3, 5, or 7, as required.
2/ Numbers indicate color symbol sequence in PIN
3/ Vertical split lamp circuit is available only by rotating base 90°

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:32Z
Check the source to verify that this is the current version before use.

Qualification inspection All applicants for qualification approval shall demonstrate that each of their items conforms to all of the requirements specified in the applicable documents singularly and in combination with all other previously qualified items, regardless of manufacturer

Group submission

a When the basic sensitive switches are qualified to MIL-S-8805, switches shall be tested in accordance with table VI.

b When the basic sensitive switches are not qualified to MIL-S-8805, the switch type for which qualification is sought shall be subjected to the complete tests in accordance with table X of MIL-S-22885

(8) TABLE VI Qualification inspection (group submission)

PIN	Inspection table of MIL-S-22885		Extent of approval
	Group	Number of samples	
M22885/96-64WW	I II III VI 1/ VII VIII	All sample units 4 (from group I) 2 (from group I) 2 (from group I) 2 (from group I) 2 each color 2/	All
M22885/96-Y1W	I II	All sample units 2 (from group I)	

1/ Sea level, resistive dc load only
2/ Two cap assemblies for each color (may be installed in switch bodies submitted to previous testing)

CONCLUDING MATERIAL

Preparing activity
Navy - EC

Agent.
DLA - ES

(Project 5930-N709)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:32Z
Check the source to verify that this is the current version before use.